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Design and Evaluation of a Legal AI Assistant: A Comparative Study of RAG and Fine-Tuning



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Context and Motivation

The rapid development of Large Language Models (LLMs) has opened new possibilities for domain-specific AI applications. However, general-purpose models such as GPT-4 often lack precision when applied to specialized fields requiring exact source citation, up-to-date legislation, and structured reasoning, as is the case in legal question answering.

French legal information is publicly available (Légifrance, EUR-Lex) yet remains difficult to access for non-expert citizens. This project addresses the following research question:

> *Can a lightweight open-source LLM, specialized through fine-tuning or retrieval augmentation, rival a large generalist model on a French legal QA task?*

This project is carried out as a collaborative effort between two students from complementary master programmes with agreement from JACQUENET François and HABRARD Amaury: M1 MLDM and M1 DSC.

Main Objectives

The project has three main objectives:

1. **Build and compare two adaptation approaches** on a lightweight open-source LLM (Mistral-7B or Phi-3 Mini):
   - **LoRA/QLoRA fine-tuning**, specializing model parameters on an annotated French legal QA dataset (LLeQA, BSARD), so the model learns to cite sources precisely and adopt the correct response format.
   - **Retrieval-Augmented Generation (RAG)**, connecting the model to a vector database of French legal texts (Légifrance), allowing it to retrieve and ground its answers in real legislation at inference time.
2. **Conduct a rigorous benchmark evaluation** comparing our systems against existing models (Mistral base, LLaMA-3, CroissantLLM, GPT-4o zero-shot and few-shot) using established metrics: ROUGE-L, BERTScore, Recall@K, and structured human evaluation. Error analysis will cover hallucination patterns, confusion between legal systems, and out-of-scope question handling.
3. **Deploy a functional web interface** exposing the best-performing system through a REST API (FastAPI) and a lightweight front-end, enabling live demonstration at the July oral presentation.

Methodology and Technical Stack

Component	Tool / Resource
Base model	Mistral-7B Instruct or Phi-3 Mini (HuggingFace)
Fine-tuning	HuggingFace `transformers` + `peft` (LoRA/QLoRA)
RAG pipeline	LangChain + ChromaDB or FAISS
Datasets	LLeQA (maastrichtlawtech/lleqa), BSARD, Légifrance API
Evaluation	ROUGE-L, BERTScore, Recall@K, human evaluation
Backend API	FastAPI (Python)
Frontend	Streamlit or React
Compute	Google Colab / Kaggle (free GPU)

Planned Schedule (12 Weeks)

Phase	Weeks	Key deliverables
Foundations	1–3	Literature review, baseline model evaluation, dataset preparation
Experimentation	4–8	LoRA fine-tuning, RAG implementation, comparative benchmark
Deployment & write-up	9–12	API + interface, full evaluation, final report and slides

Expected Outcomes and Scientific Contribution

The project will produce: (1) a fine-tuned and/or RAG-augmented legal QA system evaluated on public benchmarks (2) a comparative analysis addressing an open research question in French-language NLP and (3) a deployable web demonstration.

The broader scientific contribution lies in the under-explored intersection of lightweight LLM specialization and French legal NLP, an area where rigorous benchmarking remains limited in the current literature.